Symbolic gate atlas

Exact identities. Explicit phase policies. Native entangler counts.

6 targets × 32 basis panels

Each target has one page with 32 basis panels in alphabetical order. Blue labels denote literal matrix equality. Brown labels denote a projective circuit with its exact scalar factor. Both appear when the projective construction is strictly cheaper. Physical GP operations are forbidden.
E counts native two-qubit primitives, L counts one-qubit operations. Scores are displayed rounded; the archive stores exact expressions. Only an explicit matching certificate establishes a minimum entangler count. A verified upper bound does not establish minimum local cost. Proof conditions, source hashes and attempt histories remain in the archive.
Open cases are shown as open. Proved obstructions preserve the supplied basis and may be conditional for parameterized families. Register constructors use the recorded integer-width or matrix fixtures. These pages do not claim a solution for every register size.
Circuits are drawn by Qiskit's LaTeX drawer. Long circuits continue on successive rows in time order. An explicit $[G]^{\times n}$ box denotes n consecutive identical primitive operations; their full physical count remains in E.
Long exact expressions use numbered η_j labels defined in the expression glossary. No parameter is rounded or dropped.
The separate Clifford volume lists every exact phase in each distinct one-qubit alphabet, with 24 circuits per page. Identical alphabets shared by multiple presets are printed once.

Archive SHA256: 98f19ce6d2d1b4af05dccd65fdda73f0f49f159d08acec0476eb1d0fe6669033

Alphabetical target index

Links lead to the 32-basis page for each target.

Canonical 4
CCX 5
CP 6
CX 7
PhasedFSim 8
SWAP 9

Native basis key

Every panel preserves its supplied alphabet. Native units and macro counts matter.

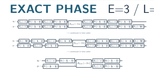
Preset	Entangler	Local alphabet	Scope / units
aqt_rxx	Rxx	R, Rz	exact continuous
berkeley_b	B	U3	exact continuous
clifford_only	CX	H, S	Clifford analysis/control; not universal.
clifford_t_cx	CX	H, S, T, Tdg	Dense universal; exact only for representable targets, not arbitrary symbolic angles.
clifford_t_cz	CZ	H, S, T, Tdg	approximate discrete
cx_h_phase	CX	H, P	exact continuous
cx_rz_ry	CX	Rz, Ry	exact continuous
cx_u	CX	U3	exact continuous
cz_rz_rx	CZ	Rz, Rx	exact continuous
cz_u	CZ	U3	exact continuous
google_cz	CZ	PhasedXZ	exact continuous
google_fsim	fSim	PhasedXZ	exact continuous
google_sqrt_iswap	SiSWAP	PhasedXZ	exact continuous
google_syc	SYC	PhasedXZ	exact continuous
ibm_cx	CX	Rz, Sx, X	exact continuous
ibm_cz	CZ	Rz, Sx, X	exact continuous
ibm_ecr	ECR	Rz, Sx, X	exact continuous
ionq_aria	IonQMS	GPi, GPi2	Native turns; interaction theta restricted to [0,1/4] on hardware.
ionq_aria_full	IonQMSMax	GPi, GPi2	exact continuous
ionq_forte	IonQZZ	GPi, GPi2	exact continuous
iqm_cz	CZ	R	R(theta,phi) represents PRX in radians; IQM API units require conversion.
iswap_u	iSWAP	U3	exact continuous
quantinuum_zzmax	ZZMax	PhasedX, Rz	exact continuous
quantinuum_zzphase	ZZPhase	PhasedX, Rz	exact continuous
rigetti_cz	CZ	Rz, Sx, X	Sx differs from native Rx(pi/2) by a tracked global phase.
sqrt_iswap_u	SiSWAP	U3	exact continuous
sqrt_swap_u	SSWAP	U3	exact continuous
tket_tk2	TK2	PhasedX, Rz	Compiler macro count; expand TK2 before comparing physical pulse counts.
xx90_u	XX90	U3	exact continuous
xx_rz_ry	Rxx	Rz, Ry	exact continuous
zx_u	Rzx	U3	exact continuous
zz_u	Rzz	U3	exact continuous

Canonical (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=41 / score ≈ 102.928 — upper bound



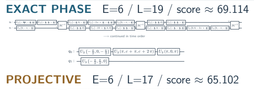
PROJECTIVE E=3 / L=20 / score ≈ 56.872 — upper bound



$U = \eta_1\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 65.102 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

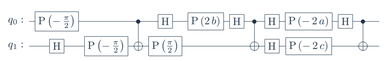
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

PROJECTIVE E=3 / L=12 / score ≈ 36.612 — upper bound



$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — upper bound



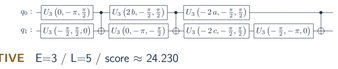
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — upper bound



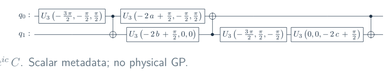
$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — upper bound



PROJECTIVE E=3 / L=5 / score ≈ 24.230 — upper bound



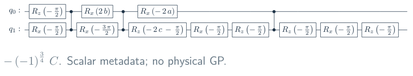
$U = e^{ic}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — upper bound



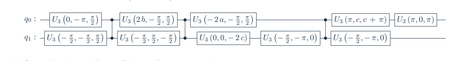
PROJECTIVE E=3 / L=11 / score ≈ 35.822 — upper bound



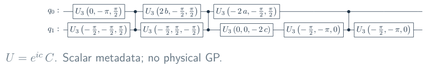
$U = -\left(-1\right)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 34.660 — upper bound



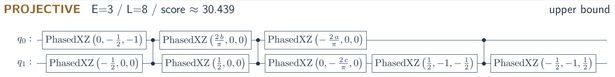
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — upper bound



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_cz

PROJECTIVE E=3 / L=8 / score ≈ 30.439 — upper bound



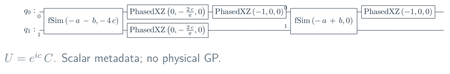
$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 21.365 — upper bound



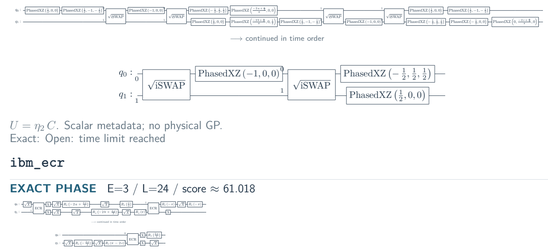
PROJECTIVE E=2 / L=4 / score ≈ 16.693 — upper bound



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

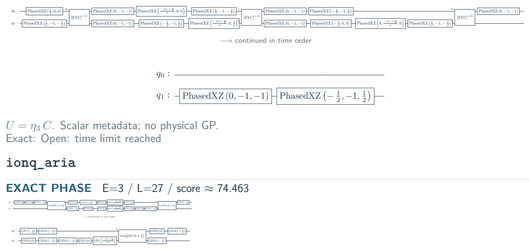
PROJECTIVE E=6 / L=17 / score ≈ 63.893 — upper bound



$U = \eta_2\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_syc

PROJECTIVE E=18 / L=17 / score ≈ 110.693 — upper bound



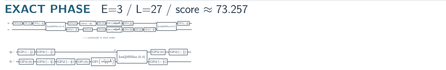
$U = \eta_3\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — upper bound



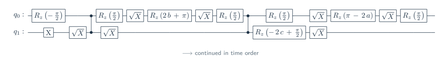
PROJECTIVE E=3 / L=13 / score ≈ 39.012 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

PROJECTIVE E=3 / L=20 / score ≈ 53.620 — upper bound



$U = -1\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — upper bound

PROJECTIVE E=3 / L=22 / score ≈ 57.214 — upper bound



$U = -\left(-1\right)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=3 / L=27 / score ≈ 74.463 — upper bound

PROJECTIVE E=3 / L=27 / score ≈ 74.263 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 73.257 — upper bound

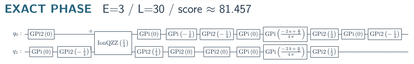
PROJECTIVE E=3 / L=27 / score ≈ 73.057 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=3 / L=30 / score ≈ 81.457 — upper bound



iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — upper bound



PROJECTIVE E=3 / L=13 / score ≈ 40.452 — upper bound



$U = -\left(-1\right)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=3 / L=13 / score ≈ 41.278 — upper bound



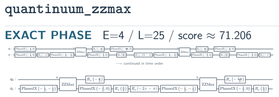
PROJECTIVE E=3 / L=10 / score ≈ 35.260 — upper bound



$U = \frac{\sqrt{2}(1+i)e^{ib}}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 71.206 — upper bound



PROJECTIVE E=3 / L=10 / score ≈ 34.293 — upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 53.049 — upper bound



rigetti_cz

PROJECTIVE E=3 / L=20 / score ≈ 53.620 — upper bound



$U = -1\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 67.914 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = -\sqrt[4]{-1}e^{ic}\, C$. Scalar metadata; no physical GP.

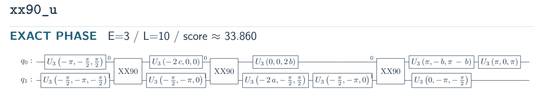
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.997 — conditional minimum



xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 33.860 — upper bound



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — upper bound



$U = ie^{-ib}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=41 / score ≈ 98.888 — upper bound



PROJECTIVE E=3 / L=31 / score ≈ 80.068 — upper bound



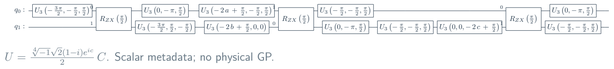
$U = \eta_1\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — upper bound



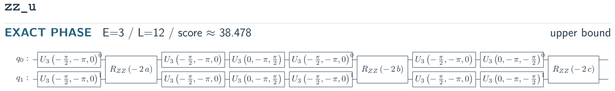
PROJECTIVE E=3 / L=11 / score ≈ 39.272 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1-i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — upper bound

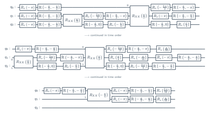


CCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

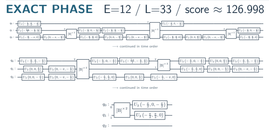
PROJECTIVE E=5 / L=40 / score ≈ 112.926 — upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 — upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

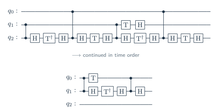
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 — proved minimum



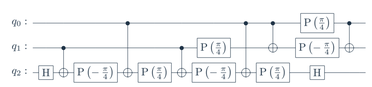
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved minimum



cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 — proved minimum



cx_rz_ry

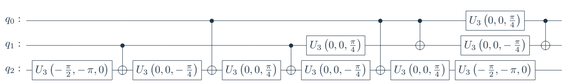
PROJECTIVE E=6 / L=10 / score ≈ 47.220 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

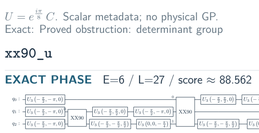
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 45.254 — proved minimum



cz_rz_rx

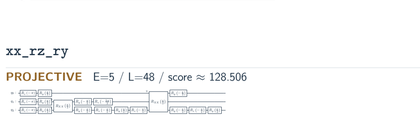
PROJECTIVE E=6 / L=40 / score ≈ 113.280 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

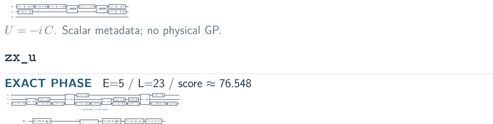
cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.726 — proved minimum



google_cz

EXACT PHASE E=6 / L=21 / score ≈ 74.526 — proved minimum



google_fsim

EXACT PHASE E=5 / L=21 / score ≈ 72.346 — upper bound



google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — upper bound

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 — upper bound

ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.218 — proved minimum

$U = e^{-\frac{3\pi i}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

PROJECTIVE E=6 / L=35 / score ≈ 98.228 — proved minimum

$U = e^{-\frac{3\pi i}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=48 / score ≈ 134.932 — upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=48 / score ≈ 132.520 — upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 149.320 — upper bound

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — proved minimum

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=33 / score ≈ 118.198 — upper bound

PROJECTIVE E=10 / L=31 / score ≈ 114.186 — upper bound

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=6 / L=35 / score ≈ 103.498 — proved minimum

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=5 / L=29 / score ≈ 89.288 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 — proved minimum

$U = e^{\frac{1\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — upper bound

PROJECTIVE E=12 / L=33 / score ≈ 124.598 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=5 / L=32 / score ≈ 95.322 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 — proved minimum

xx_rz_ry

PROJECTIVE E=5 / L=48 / score ≈ 128.506 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=23 / score ≈ 76.548 — upper bound

PROJECTIVE E=5 / L=21 / score ≈ 72.536 — upper bound

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=25 / score ≈ 80.160 — upper bound

PROJECTIVE E=5 / L=23 / score ≈ 76.148 — upper bound

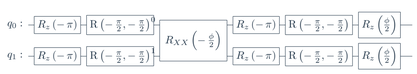
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

CP (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

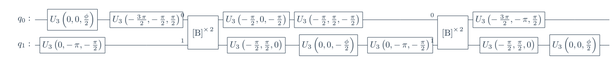
PROJECTIVE E=1 / L=10 / score ≈ 26.230 conditional minimum



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.866 upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

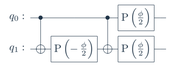
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

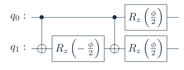
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 upper bound



cx_rz_ry

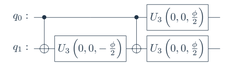
PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 upper bound



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 25.616 upper bound



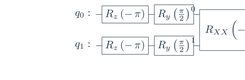
$U = -e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 upper bound



PROJECTIVE E=2 / L=5 / score ≈ 19.430 upper bound



$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.421 upper bound



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 conditional minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 42.057 upper bound



PROJECTIVE E=2 / L=6 / score ≈ 21.036 upper bound

$U = ie^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.257 upper bound



ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound



$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 upper bound

$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 45.635 upper bound

$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.831 upper bound

$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.601 conditional minimum

$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 upper bound

$U = -e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 conditional minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.036 conditional minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.950 upper bound

$U = -ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.936 conditional minimum

$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound

$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.066 upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.066 upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.940 conditional minimum

$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.036 upper bound

$U = ie^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 conditional minimum

$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.026 conditional minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.614 conditional minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

CX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

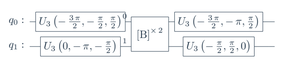
PROJECTIVE E=1 / L=5 / score ≈ 15.218 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — upper bound



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



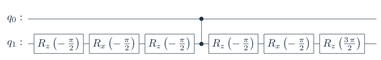
cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 17.212 — proved minimum



cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved minimum



google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 17.824 — upper bound



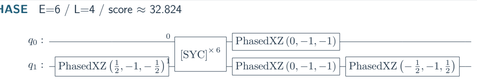
PROJECTIVE E=2 / L=4 / score ≈ 17.224 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 — upper bound



ibm_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



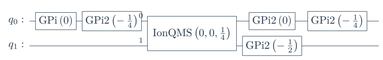
ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 — proved minimum



ionq_aria

PROJECTIVE E=1 / L=5 / score ≈ 15.416 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

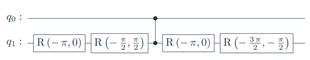
PROJECTIVE E=1 / L=6 / score ≈ 17.814 — upper bound



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

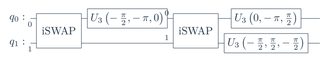
iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved minimum



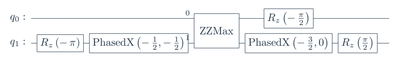
iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved minimum



quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=5 / score ≈ 15.216 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

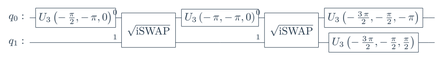
rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — upper bound



sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — upper bound



PROJECTIVE E=2 / L=4 / score ≈ 17.224 — upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

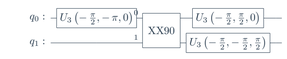
PROJECTIVE E=1 / L=5 / score ≈ 15.222 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved minimum



xx_rz_ry

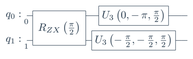
PROJECTIVE E=1 / L=7 / score ≈ 19.416 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

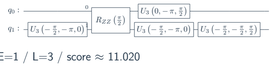
zx_u

EXACT PHASE E=1 / L=2 / score ≈ 9.014 — upper bound



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — upper bound



PROJECTIVE E=1 / L=3 / score ≈ 11.020 — upper bound



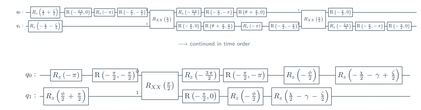
$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

PhasedFSim $(\theta, \zeta, \chi, \gamma, \phi)$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

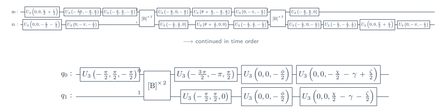
PROJECTIVE E=3 / L=26 / score ≈ 70.084 — upper bound



$U = \eta_R\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=23 / score ≈ 78.338 — upper bound



$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

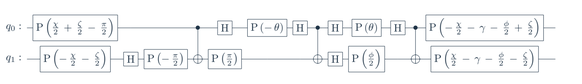
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

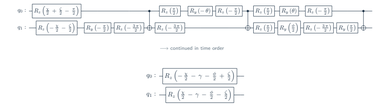
EXACT PHASE E=3 / L=15 / score ≈ 43.418 — upper bound



$U = \eta_{11}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_rz_ry

PROJECTIVE E=3 / L=17 / score ≈ 49.034 — upper bound



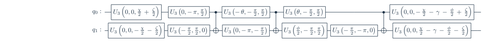
$U = \eta_T\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=11 / score ≈ 37.466 — upper bound



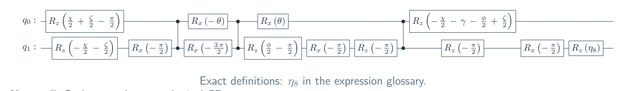
PROJECTIVE E=3 / L=11 / score ≈ 37.266 — upper bound



$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=3 / L=13 / score ≈ 40.226 — upper bound

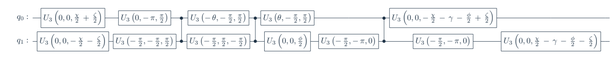


Exact definitions: η_8 in the expression glossary.

$U = \eta_T\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

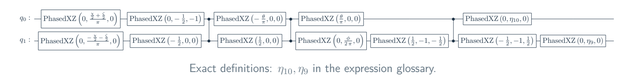
cz_u

EXACT PHASE E=3 / L=12 / score ≈ 39.672 — upper bound



google_cz

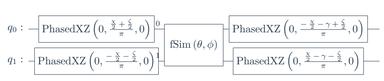
EXACT PHASE E=3 / L=12 / score ≈ 40.784 — upper bound



Exact definitions: η_{10}, η_9 in the expression glossary.

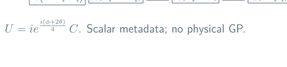
google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 — upper bound



google_sqrt_iswap

EXACT PHASE E=6 / L=25 / score ≈ 84.383 — upper bound



PROJECTIVE E=6 / L=23 / score ≈ 79.111 — upper bound

$U = -\frac{\sqrt{2}(1-i)e^{-i\theta}}{2}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=25 / score ≈ 130.583 — upper bound



PROJECTIVE E=18 / L=23 / score ≈ 125.911 — upper bound

$U = -ie^{-i\theta}\, C$. Scalar metadata; no physical GP.

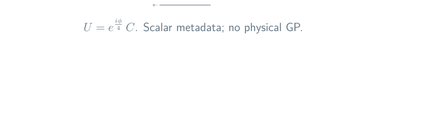
ibm_cx

PROJECTIVE E=3 / L=19 / score ≈ 52.224 — upper bound

$U = -e^{-\frac{i(\phi_3+\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — upper bound



Exact definitions: η_{11} in the expression glossary.

$U = \sqrt[4]{-1}e^{-\frac{i(\phi_3+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=27 / score ≈ 68.224 — upper bound

$U = -ie^{-\frac{i(4+\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=39 / score ≈ 103.869 — upper bound

$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=39 / score ≈ 102.663 — upper bound

$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=42 / score ≈ 111.063 — upper bound

$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=25 / score ≈ 68.700 — upper bound

$U = \eta_T\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=17 / score ≈ 51.102 — upper bound

PROJECTIVE E=3 / L=14 / score ≈ 44.284 — upper bound

$U = \eta_{13}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=13 / score ≈ 41.299 — upper bound

$U = -ie^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=3 / L=22 / score ≈ 62.057 — upper bound

$U = e^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — upper bound

Exact definitions: η_{11} in the expression glossary.

$U = \sqrt[4]{-1}e^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.138 — upper bound

sqrt_swap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.938 — upper bound

PROJECTIVE E=6 / L=23 / score ≈ 77.138 — upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=4 / score ≈ 14.005 — conditional minimum

$U = e^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=14 / score ≈ 43.684 — upper bound

PROJECTIVE E=3 / L=12 / score ≈ 38.472 — upper bound

$U = ie^{\frac{i(\phi+2\theta)}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=36 / score ≈ 91.078 — upper bound

$U = \eta_6\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=17 / score ≈ 52.508 — upper bound

zz_u

EXACT PHASE E=3 / L=18 / score ≈ 52.314 — upper bound

PROJECTIVE E=3 / L=16 / score ≈ 47.502 — upper bound

$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

SWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

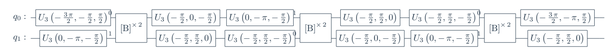
PROJECTIVE E=3 / L=15 / score ≈ 45.654 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=12 / score ≈ 52.872 — upper bound



clifford_only

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



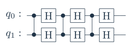
clifford_t_cx

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



clifford_t_cz

EXACT PHASE E=3 / L=6 / score ≈ 24.000 — proved minimum



cx_h_phase

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



cx_rz_ry

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



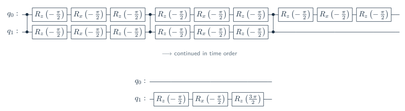
cx_u

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



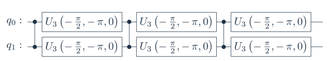
cz_rz_rx

EXACT PHASE E=3 / L=18 / score ≈ 51.636 — proved minimum



cz_u

EXACT PHASE E=3 / L=6 / score ≈ 25.236 — proved minimum



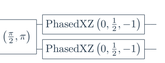
google_cz

EXACT PHASE E=3 / L=6 / score ≈ 26.436 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.616 — proved minimum



google_sqrt_iswap

EXACT PHASE E=6 / L=12 / score ≈ 52.272 — upper bound



PROJECTIVE E=6 / L=12 / score ≈ 51.672 — upper bound

$U = \frac{\sqrt{2}(1-i)^3}{4}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=12 / score ≈ 98.472 — upper bound



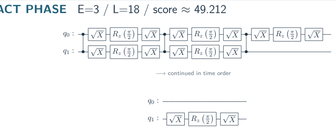
ibm_cx

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



ibm_cz

EXACT PHASE E=3 / L=18 / score ≈ 49.212 — proved minimum



ibm_ecr

EXACT PHASE E=3 / L=12 / score ≈ 36.606 — proved minimum



PROJECTIVE E=3 / L=11 / score ≈ 34.810 — proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=3 / L=15 / score ≈ 46.248 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

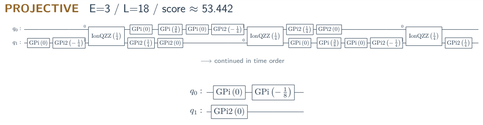
PROJECTIVE E=3 / L=15 / score ≈ 45.042 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

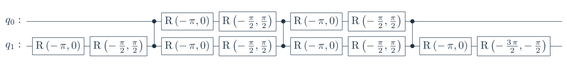
PROJECTIVE E=3 / L=18 / score ≈ 53.442 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=3 / L=12 / score ≈ 38.448 — proved minimum



iswap_u

EXACT PHASE E=3 / L=5 / score ≈ 23.430 — proved minimum



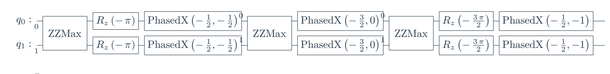
PROJECTIVE E=3 / L=4 / score ≈ 20.624 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

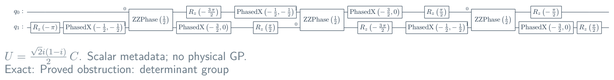
PROJECTIVE E=3 / L=10 / score ≈ 34.032 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

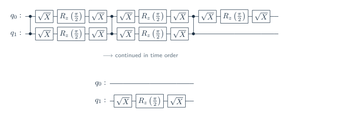
PROJECTIVE E=3 / L=13 / score ≈ 41.644 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=3 / L=18 / score ≈ 49.212 — proved minimum



sqrt_iswap_u

EXACT PHASE E=6 / L=12 / score ≈ 51.672 — upper bound



sqrt_swap_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



tket_tk2

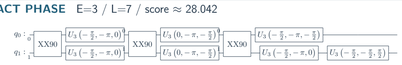
PROJECTIVE E=1 / L=0 / score ≈ 4.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=7 / score ≈ 28.042 — proved minimum



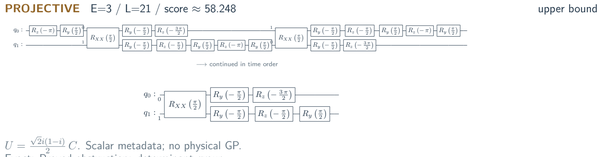
PROJECTIVE E=3 / L=6 / score ≈ 25.636 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

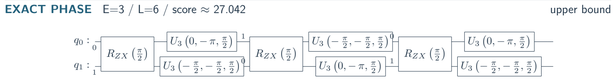
PROJECTIVE E=3 / L=21 / score ≈ 58.248 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

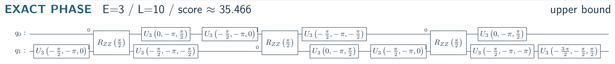
zx_u

EXACT PHASE E=3 / L=6 / score ≈ 27.042 — upper bound



zz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.466 — upper bound



PROJECTIVE E=3 / L=9 / score ≈ 33.060 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{1} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1+i)^{3}}{4}$$

$$\eta_{2} = \frac{\sqrt{2}(-1+i)e^{i(a+b+c)}}{2}$$

$$\eta_{3} = \sin\left(a+b+c\right) - i\cos\left(a+b+c\right)$$

$$\eta_{4} = \pi - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{5} = -\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + 2\pi$$

$$\eta_{6} = \frac{\sqrt{2}i(1+i)^{3}e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{7} = -\left(-1\right)^{\frac{3}{4}}e^{-\frac{i(4\chi+\phi)}{4}}$$

$$\eta_{8} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{\pi}{2}$$

$$\eta_{9} = \frac{\frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2}}{\pi}$$

$$\eta_{10} = \frac{-\frac{\chi}{2} - \gamma - \frac{\phi}{2} + \frac{\zeta}{2}}{\pi}$$

$$\eta_{11} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{3\pi}{2}$$

$$\eta_{12} = \frac{\sqrt{2}(1+i)^{3}e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{13} = \frac{\sqrt{2}i(1+i)e^{\frac{i(\phi-2\theta)}{4}}}{2}$$